Linear Regression with NumPy

Introduction

In this document, we implement a simple linear regression model using NumPy. Linear regression is a fundamental technique in machine learning and statistics used to model the relationship between a dependent variable and an independent variable. We will follow these steps:

1. Create a dataset.
2. Define the linear regression model.
3. Describe the cost function (Mean Squared Error - MSE).
4. Implement gradient descent for optimization.

1 Dataset Representation

We create a dataset (x, y) with m rows, $n = 1$ feature x and y as the target variable.

x	y
$x^{(1)}$	$y^{(1)}$
$x^{(2)}$	$y^{(2)}$
$\vdots$	$\vdots$
$x^{(m)}$	$y^{(m)}$

2 Linear Regression Model

The model is represented by a function f defined as :

$$f(x) = ax + b \tag{1}$$

In matrix form, we have :

$$F = X\theta \tag{2}$$

where:

$$F = \begin{bmatrix} f(x^{(1)}) \\ f(x^{(2)}) \\ \vdots \\ f(x^{(m)}) \end{bmatrix} \quad ; \ \dim(F) = (m \times 1) \tag{3}$$

$$X = \begin{bmatrix} x^{(1)} & 1 \\ x^{(2)} & 1 \\ \vdots & \vdots \\ x^{(m)} & 1 \end{bmatrix} \quad ; \ \dim(X) = (m \times 2) \tag{4}$$

$$\theta = \begin{bmatrix} a \\ b \end{bmatrix} \quad ; \ \dim(\theta) = (2 \times 1) \tag{5}$$

3 Cost Function

The cost function J is given by the Mean Squared Error (MSE):

$$J(a,b) = \frac{1}{2m} \sum_{i=1}^{m} \left(ax^{(i)} + b - y^{(i)} \right)^2 \tag{6}$$

In matrix form:

$$J(\theta) = \frac{1}{2m} \sum X^T (X\theta - Y)^2 \tag{7}$$

where:

$$Y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix} \tag{8}$$

and

$$X^T = \begin{bmatrix} x^{(1)} & x^{(2)} & ... & x^{(m)} \\ 1 & 1 & ... & 1 \end{bmatrix} \tag{9}$$

4 Gradient Descent

Gradient Descent is an optimization algorithm used to minimize $J(a,b)$ by iteratively updating a and b:

$$a_{i+1} = a_i - \alpha \frac{\partial J}{\partial a} \tag{10}$$

$$b_{i+1} = b_i - \alpha \frac{\partial J}{\partial b} \tag{11}$$

where α is the learning rate ($\alpha > 0$). The gradients are computed as:

$$\begin{aligned} \frac{\partial J(a,b)}{\partial a} &= \frac{1}{2m} \sum_{i=1}^{m} \frac{\partial}{\partial a} \left(ax^{(i)} + b - y^{(i)} \right)^2 && (12) \\ &= \frac{1}{2m} \sum_{i=1}^{m} 2x^{(i)} \left(ax^{(i)} + b - y^{(i)} \right) && (13) \\ &= \frac{1}{m} \sum_{i=1}^{m} \left(ax^{(i)} + b - y^{(i)} \right) x^{(i)} && (14) \end{aligned}$$

$$\begin{aligned} \frac{\partial J(a,b)}{\partial b} &= \frac{1}{2m} \sum_{i=1}^{m} \frac{\partial}{\partial b} \left(ax^{(i)} + b - y^{(i)} \right)^2 && (15) \\ &= \frac{1}{2m} \sum_{i=1}^{m} 2 \left(ax^{(i)} + b - y^{(i)} \right) && (16) \\ &= \frac{1}{m} \sum_{i=1}^{m} \left(ax^{(i)} + b - y^{(i)} \right) && (17) \end{aligned}$$

In matrix form:

$$\begin{aligned} \frac{\partial J(\theta)}{\partial \theta} &= \begin{bmatrix} \frac{\partial J(a,b)}{\partial a} \\ \frac{\partial J(a,b)}{\partial b} \end{bmatrix} && (18) \\ &= \frac{1}{m} X^T (X\theta - Y) && (19) \end{aligned}$$